

```
project/
├── controllers/
│   └── userController.js
├── middlewares/
│   └── upload.js
├── models/
│   └── User.js
├── routes/
│   └── userRoutes.js
├── uploads/
│   └── users/
├── server.js
└── .env
```

npm install multer

1 Cour

```
// models/User.js
const mongoose = require("mongoose");
const courSchema = new mongoose.Schema({

  titre: String,
  description: String,
  niveau: String,
  categorie: String,
  image: String,

}, { timestamps: true });
module.exports = mongoose.model("Cour", courSchema);
```

2 Multer Upload Middleware – middlewares/upload.js

```
const multer = require("multer");
const path = require("path");
```

```

const fs = require("fs");

// Create folder if not exists
const uploadDir = "uploads/";
if (!fs.existsSync(uploadDir)) {
  fs.mkdirSync(uploadDir, { recursive: true });
}

const storage = multer.diskStorage({
  destination: (req, file, cb) => {
    cb(null, uploadDir);
  },
  filename: (req, file, cb) => {
    const uniqueName =
      Date.now() + "-" + Math.round(Math.random() * 1e9);
    cb(null, uniqueName + path.extname(file.originalname));
  },
});

const fileFilter = (req, file, cb) => {
  if (file.mimetype.startsWith("image/")) {
    cb(null, true);
  } else {
    cb(new Error("Only image files are allowed!"), false);
  }
};

const upload = multer({
  storage,
  limits: { fileSize: 2 * 1024 * 1024 }, // 2MB
  fileFilter,
});

module.exports = upload;

```

③ User Controller – controllers/userController.js

```

// controllers/userController.js
const Cour = require("../models/Cour");
// Ajouter un utilisateur (admin uniquement)

```

```

exports.ajouterCour = async (req, res) => {
  try { const { titre, description, niveau, categorie } = req.body;
  const nouvelCour = new Cour({
    titre,
    description,
    niveau,
    categorie,
    image: req.file ? req.file.filename : null
  });
  await nouvelCour.save();
  res.status(201).json(nouvelCour);
} catch (err) {
  res.status(400).json({ message: "Erreur d'ajout", error: err.message
});
}
};

// Récupérer tous les utilisateurs
exports.listerCour = async (req, res) => {
  try {
    const cours = await Cour.find();
    res.json(cours);
  } catch (err) {
    res.status(500).json({ error: err.message });
  }
};

```

4 Routes – routes/coursRoutes.js

```

const express = require("express");
const router = express.Router();
const userController = require("../controllers/userController");
const protect = require("../middleware/authMiddleware");
const authorize = require("../middleware/roleMiddleware");

router.post("/ajouter", userController.ajouterUtilisateur);
router.get("/", protect, authorize(["baybsitter", "admin"])
, userController.listerUtilisateurs);
module.exports = router;

```

5 Server – server.js

```
const express = require("express");
const mongoose = require("mongoose");
const dotenv = require("dotenv");
const userRoutes = require("./routes/userRoutes");
const path = require("path");
```

```
dotenv.config();
```

```
const app = express();
```

```
// Middlewares
```

```
app.use(express.json());
```

```
app.use(express.urlencoded({ extended: true }));
```

```
// Static folder for images
```

```
app.use("/uploads", express.static(path.join(path.resolve(),
"/uploads")));
```

```
// Routes
```

```
app.use("/api", userRoutes);
```

```
// MongoDB connection
```

```
mongoose
```

```
.connect(process.env.MONGO_URI)
```

```
.then(() => {
```

```
  console.log("MongoDB connected");
```

```
  app.listen(5000, () => {
```

```
    console.log("Server running on port 5000");
```

```
  });
```

```
})
```

```
.catch((err) => {
```

```
  console.error(err);
```

```
});
```